Further Algebra

Questions 1–153

1.16 — Question 1.

a) Sketch the graph of

$$y = (x-2)(x+1)(x+6)$$

b) Find the values of x, in set notation, where:
   i) $y > 0$
   ii) $y \leq 0$

c) Find the values of x, in interval notation, where:
   i) $y < 0$
   ii) $y \geq 0$

1.16 — Question 2. Solve the following inequalities, writing your answers in set notation:

a) $(x+1)(x+2)(x+4) > 0$
b) $(x-2)(x-3)(x+5) < 0$
c) $(2x+1)(x+4)(x-8) \geq 0$
d) $(3x-2)(2x-5)(x+2) \leq 0$
e) $(2x-1)(x^2+9x+20) > 0$
f) $(2x-3)^2(x+1) > 0$
g) $(5x+2)x^2 \leq 0$

1.16 — Question 3. Solve the following inequalities, writing your answers in set notation:

a) $(1-x)(x+5)(3+x) > 0$
b) $(x-7)(5-x)(x+1) < 0$
c) $(3x+4)(x-2)(1-x) \geq 0$
d) $(2-3x)(2x+1)(4-5x) \leq 0$
e) $(3-2x)(x^2-x-2) < 0$
f) $(4-5x)(3-x)^2 > 0$
g) $2x^2(4-7x) \leq 0$

1.16 — Question 4. Solve the following inequalities, writing your answers in interval notation:

a) $x^3+5x^2-18x-72 > 0$
b) $4x^3-8x^2+3x \leq 0$
c) $9x^3-3x^2-5x-1 \geq 0$
d) $4+28x-9x^2-63x^3 < 0$
e) $6x^2-5x^3 \geq 0$
f) $8+61x+104x^2-48x^3 < 0$

1.16 — Question 5. Solve the following inequalities, writing your answers in set notation:

a) $(1+x)(3-x)(x+5) \geq (1+x)(x-3)$
b) $(2+x)(x-4)x \leq (4-x)x^2$
c) $(5-x)(3+x) > (3+x)^2(x-5)$
d) $(4-3x)^2(6-5x) < (3x-4)(5x-6)^2$

1.16 — Question 6. Given that $k > 3$, solve the following inequalities, writing your answers in set notation:

a) $(x-1)(x+1)(x-k) > 0$
b) $(x+3)(2x+5)(2x-k) < 0$
c) $(x+k)(2x+k)(x-2k) \geq 0$
d) $(kx+4)(kx-3)(3x+5) \leq 0$

1.16 — Question 7. It is given that the inequality

$$(6x^2 + 19x + p)(x + 2) \geq 0$$

has the solution

$$\left\{x : x \in \mathbb{R}, -\frac{7}{2} \leq x \leq -2\right\} \cup \{x : x \in \mathbb{R}, x \geq q\}$$

where p and q are integers.
Find the values of p and q

1.16 — Question 8. It is given that the inequality

$$px^3 + 17x^2 + qx - 72 \leq 0$$

has the solution $x \in (-\infty, -4] \cup \left[-\frac{9}{4}, r\right]$ where p, q and r are integers.
Find the values of p, q and r

1.16 — Question 9. It is given that $f(x) = (3 + 2x)(2 + x)^2$

a) Solve $f(x) > 0$, writing your answer in interval notation.

$y = f(x)$ is mapped to $y = g(x)$ by a reflection in the x-axis.

b) Solve $g(x) > 0$, writing your answer in interval notation.

Quartic Inequalities

1.16 — Question 10. Solve the following inequalities, writing your answers in set notation:

a) $(x - 1)(x - 2)(x + 3)(x + 5) > 0$

b) $(x + 4)(x + 3)(x + 2)x \leq 0$

c) $(2x - 3)(3x + 5)(x^2 - 1) \geq 0$

d) $(x^2 - 4)(9x^2 - 1) < 0$

e) $(4x^2 + 8x - 5)(3x^2 - 10x + 3) > 0$

f) $(x - 6)(x - 7)(2x + 1)^2 \geq 0$

g) $(2x + 7)^2(3x + 8)^2 > 0$

h) $(3x - 2)^3(x + 3) < 0$

1.16 — Question 11. Solve the following inequalities, writing your answers in set notation:

a) $(2-x)(x+1)(x-3)(x+5) > 0$
b) $(4+x)(5-x)(x+3)(x-2) \leq 0$
c) $(4x^2-1)(1-9x^2) < 0$
d) $(1+x-2x^2)(9x^2-6x-8) \leq 0$
e) $(3x+1)^2(2-x)(2x-1) > 0$
f) $(3-x)^2(4-3x)^2 > 0$
g) $(7x+3)^3(4x+5) \geq 0$

1.16 — Question 12. Solve the following inequalities, writing your answers in interval notation:

a) $x^4-5x^3-13x^2+53x+60 \geq 0$
b) $4x^4+28x^3+51x^2+27x < 0$
c) $259x^2+32x-48-2x^3-16x^4 \leq 0$
d) $15x^4+13x^3-6x^2 \geq 0$
e) $30x^2+325x+375-36x^3-8x^4 < 0$

1.16 — Question 13. Solve the following inequalities, writing your answers in set notation:

a) $(x+2)(x-3)(x+1)(x-5) \geq (x+2)^2(x-3)(x+1)$

b) $(5+x)(x+6)(3-x)x \leq (x-3)^2x^2$

c) $(3-x)(2x+1)(4+x)^2 > (x-3)(2x+1)^2(4+x)$

d) $(7-2x)^3(4+9x) < (2x-7)^2(4+9x)$

1.16 — Question 14. Given that $k > 3$, solve the following inequalities, writing your answers in set notation:

a) $(x-2)(x+3)(x+k)x > 0$

b) $(2x+1)(x-1)(3x-k)(2x+3) < 0$

c) $(x-k)(2x-k)(x-2k)(x+k) \geq 0$

d) $(kx-5)(kx+3)(x-2)(x+2) \leq 0$

1.16 — Question 15. It is given that the inequality

$$(px^2 - 4x + q)(8x^2 + 6x - 9) < 0$$

has the solution

$$\left\{x : x \in \mathbb{R}, -\frac{3}{2} < x < -\frac{2}{5}\right\} \cup \left\{x : x \in \mathbb{R}, r < x < \frac{3}{4}\right\}$$

where p is an integer, and q and r are rational numbers. Find the values of p, q and r

1.16 — Question 16. It is given that the inequality

$$px^4 - 12x^3 - 49x^2 + 87x + q < 0$$

has the solution $x \in \left(-\frac{5}{2}, -\frac{3}{2}\right) \cup (r, 4)$ where p, q and r are integers.
Find the values of p, q and r

1.16 — Question 17. It is given that $f(x) = (5 - 3x)(2 + x)(x - 3)^2$

a) Solve $f(x) \leq 0$, writing your answer in interval notation.

$y = f(x)$ is mapped to $y = g(x)$ by a translation by the vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$.

b) Solve $g(x) \leq 0$, writing your answer in interval notation.

1.16 — Question 18. It is given that $f(x) = (3x - 1)^3(2 - 5x)$

a) Solve $f(x) \geq 0$, writing your answer in interval notation.

$y = f(x)$ is mapped to $y = g(x)$ by a translation by the vector $\begin{pmatrix} -5 \\ 0 \end{pmatrix}$, followed by a reflection in the x-axis.

b) Solve $g(x) \geq 0$, writing your answer in interval notation.

1.16 — Question 19. Solve the following inequalities, writing your answers in set notation:

a) $x^4 - 13x^2 + 36 > 0$

b) $4x^4 - 21x^2 - 25 \leq 0$

Hidden Polynomial Inequalities

1.16 — Question 20. Solve the following inequalities, writing your answers in set notation:

a) $e^{3x} - 12e^{2x} + 41e^x - 30 \leq 0$

b) $e^{6x} - 6e^{4x} - e^{2x} + 30 > 0$

c) $e^{4x} - 4e^{3x} - 25e^{2x} + 88e^x - 60 \geq 0$

d) $4e^{8x} - 8e^{6x} - 59e^{4x} + 63e^{2x} + 180 < 0$

1.16 — Question 21. Solve the inequality

$$2^{12x} - 10 \times 2^{9x} + 160 \times 2^{3x} - 256 \leq 0$$

writing your answer in interval notation.

1.16 — Question 22. The inequality

$$(2\sin x - 1)(\sin x + 3)(\sin x + k) > 0$$

where $0 < x < 2\pi$ and k is a constant, has the solution $\left\{x : x \in \mathbb{R}, \frac{\pi}{6} < x < \frac{5\pi}{6}\right\}$
Find the possible values of k

1.16 — Question 23. Solve the following inequalities, writing your answers in set notation:

a) $x^6 - 14x^4 + 49x^2 - 36 > 0$

b) $x^6 - 9x^3 + 8 \leq 0$

c) $x^6 - 9x^4 - 8x^3 + 72x > 0$

1.17 — Question 1.

a) Sketch the graph of

$$y = \frac{x - 4}{x}$$

b) Find the values of x, in set notation, where:

i) $y > 0$

ii) $y \leq 0$

c) Find the values of x, in interval notation, where:

i) $y < 0$

ii) $y \geq 0$

1.17 — Question 2. Solve the following inequalities, writing your answers in set notation:

a) $\frac{1}{x} > 0$
b) $\frac{1}{x-3} < 0$
c) $\frac{2}{x+1} \geq 0$
d) $\frac{5}{2x+3} \leq 0$
e) $\frac{8}{4-7x} > 0$

1.17 — Question 3. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{1}{x} > 2$
b) $\frac{1}{x-2} < 3$
c) $\frac{3}{x+4} \geq -5$
d) $\frac{4}{3-2x} \leq 7$
e) $\frac{10}{6-3x} > 9$

1.17 — Question 4. Francesca is attempting to solve the following inequality:

$$\frac{5}{x-3} \geq 9 - x$$

Her working is shown below:

$$\begin{aligned} &\frac{5}{x-3} \geq 9 - x \\ \Rightarrow\ &5 \geq (9-x)(x-3) \\ \Rightarrow\ &5 \geq -27 + 12x - x^2 \\ \Rightarrow\ &x^2 - 12x + 32 \geq 0 \\ \Rightarrow\ &(x-4)(x-8) \geq 0 \\ \Rightarrow\ &x \in (-\infty, 4) \cup (8, \infty) \end{aligned}$$

a) Identify the error Francesca has made.
b) Write out a correct solution for his problem, giving your answer in interval notation.

1.17 — Question 5. Solve the following inequalities, writing your answers in set notation:
a) $\frac{x+1}{x} > 0$
b) $\frac{x-1}{2x} < 0$
c) $\frac{2x-3}{4x} \geq 0$
d) $\frac{5-3x}{6x} \leq 0$

1.17 — Question 6. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{x+2}{x} > 1$
b) $\frac{x-4}{x} < 1$
c) $\frac{2x+1}{2x} \geq 2$
d) $\frac{5-3x}{x} \leq -4$

1.17 — Question 7. Solve the following inequalities, writing your answers in set notation:

a) $\frac{x}{x+1} > 0$
b) $\frac{2x}{x-3} < 0$
c) $\frac{4x}{2x+3} \geq 0$
d) $\frac{3x}{3-x} \leq 0$

1.17 — Question 8. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{x-3}{x+1} > 0$

b) $\frac{x+4}{x-5} < 0$

c) $\frac{2x+1}{x+3} \geq 0$

d) $\frac{5-x}{2x+5} \leq 0$

1.17 — Question 9. Solve the following inequalities, writing your answers in set notation:

a) $\frac{x+3}{x-2} > 1$

b) $\frac{x-6}{x+1} < 1$

c) $\frac{x+4}{2x-1} \geq -3$

d) $\frac{7-x}{x-8} \leq 5$

1.17 — Question 10. Solve the following inequalities, writing your answers in set notation:

a) $0 < \frac{2x+5}{x-5} < 1$

b) $-1 < \frac{3x+1}{2-x} < 1$

c) $-2 \leq \frac{5-x}{4x+1} \leq 2$

1.17 — Question 11. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{2}{x} < 2x - 3$

b) $\frac{5}{2x+1} \leq 3x - 2$

c) $\frac{x+3}{x} > x - 1$

d) $\frac{x-5}{2x} < 2 - x$

e) $\frac{x}{x-4} \geq 2x + 1$

f) $\frac{3x}{5-2x} \leq 2x - 3$

g) $\frac{x+2}{x+6} > x - 5$

h) $\frac{4-x}{6-5x} \geq 2x - 1$

i) $\frac{3-2x}{2-3x} \leq 6 - x$

1.17 — Question 12. Solve the following inequalities, writing your answers in set notation:

a) $\frac{(x-1)(x-2)}{x} > 0$

b) $\frac{(x+3)(x-5)}{x} < 0$

c) $\frac{(2x-1)(3x+4)}{2x} \leq 0$

d) $\frac{(3-x)(1+2x)}{4x} \geq 0$

1.17 — Question 13. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{x-3}{(x+1)(x+2)} > 0$

b) $\frac{x+4}{(x-2)(x-5)} < 0$

c) $\frac{2x+1}{(x+3)(x-4)} \geq 0$

d) $\frac{3-5x}{(2x+1)(x+3)} \leq 0$

1.17 — Question 14. Solve the following inequalities, writing your answers in set notation:

a) $\frac{(x-4)(x-3)}{x+2} > 0$

b) $\frac{(x+1)(x+5)}{x-3} < 0$

c) $\frac{(2x-3)(x+2)}{x-1} \geq 0$

d) $\frac{(4-x)(x-6)}{2x+3} \leq 0$

1.17 — Question 15. It is given that

$$\frac{x-3}{x+5} > x+12$$

a) Show that

$$\frac{(x+9)(x+k)}{x+5} < 0$$

where k is an integer to be found.

b) Hence solve the inequality, writing your answer in interval notation.

1.17 — Question 16. Solve the following inequalities, writing your answers in set notation:

a) $\frac{(3x-1)(x+1)}{x+2} \geq 12$

b) $\frac{(8-x)(2x+5)}{3x+1} \leq 4$

c) $\frac{(2-5x)(4-x)}{6-x} < -4$

1.17 — Question 17. Given that $\frac{x^2-9}{5x-2} \leq 3 - x$

a) Show that

$$\frac{(px+q)(x-3)}{5x-2} \leq 0$$

where p and q are integers to be found.

b) Hence solve the inequality, writing your answer in interval notation.

1.17 — Question 18. Find the values of the constants a, b and c, where $a < b$, given that the inequality

$$\frac{(x-a)(x-b)}{x-c} \geq 0$$

has the following solutions:

a) $x \in [2, 6] \cup (8, \infty)$

b) $x \in [-3, 2) \cup [5, \infty)$

1.17 — Question 19. Find the values of the constants a, b and c, where $a > b$, given that the inequality

$$\frac{(a-x)(x-b)}{x-c} \geq 0$$

has the following solutions:

a) $x \in (-\infty, -4) \cup [-2, 4]$

b) $x \in (-\infty, 5] \cup (6, 7]$

1.17 — Question 20. Solve the following inequalities, writing your answers in interval notation:

a) $\frac{(x+1)(x+2)}{(x+3)(x+4)} < 0$

b) $\frac{(x+2)(4-x)}{(x-1)(x+3)} \geq 0$

c) $\frac{(x-3)(5-x)}{(2-x)(6-x)} \leq 0$

1.17 — Question 21. By solving a relevant inequality, show that the solution to the equation $e^x + 2 = \frac{x+5}{2x-2}$ lies in the interval $1 < x < 3$

1.17 — Question 22. Given that k is a constant where $k > 5$, solve the following inequalities, writing your answers in set notation:

a) $\frac{2x}{x-k} > x + k$

b) $\frac{x-4}{k-x} < 4 - x$

c) $\frac{x^2-k^2}{x+3} \geq k - x$

1.18 — Question 1.

Sketch the following curves, identifying any intersections with the axes and any asymptotes clearly:

a) $y = \frac{x-1}{x}$

b) $y = \frac{x-5}{x+1}$

c) $y = \frac{x+2}{x+3}$

d) $y = \frac{x-2}{x-4}$

e) $y = \frac{2-x}{1-x}$

f) $y = \frac{3-x}{5-x}$

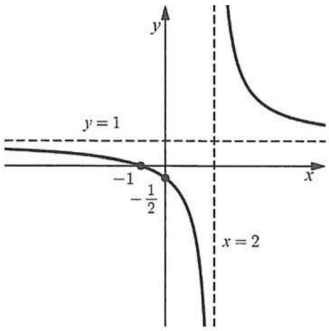
1.18 — Question 2.

Each of the following graphs are curves of the form

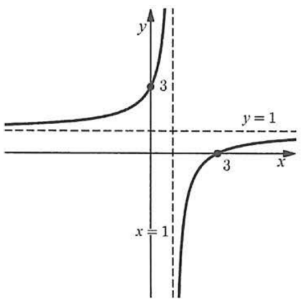
$$y = \frac{x+a}{x+b}$$

where a and b are integer constants to be found. In each case find the equation of the curve.

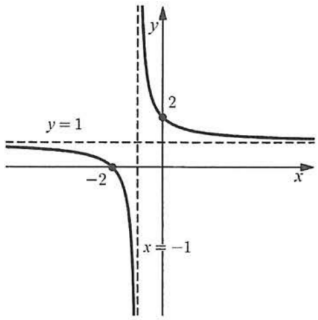
a)



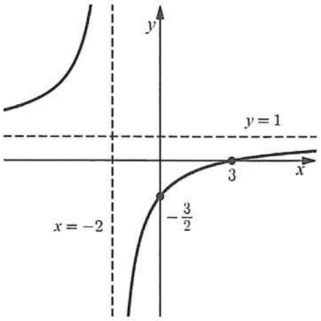
b)



c)



d)



1.18 — Question 3.

State the asymptotes for each of the graphs of the following curves:

a) $y = \frac{3x+1}{x-2}$
b) $y = \frac{4x-1}{2x+1}$
c) $y = \frac{5x+3}{x+2}$
d) $y = \frac{2x-1}{3x+5}$
e) $y = \frac{x+2}{5x-1}$
f) $y = \frac{ax+b}{cx+d}$ for constants, a, b, c and d

1.18 — Question 4.

Sketch the following curves, identifying any intersections with the axes and any asymptotes clearly:

a) $y = \frac{2x-1}{x+3}$
b) $y = \frac{3x+1}{x+2}$
c) $y = \frac{5x-3}{x-1}$
d) $y = \frac{4x-2}{x+7}$
e) $y = \frac{x+4}{2x-5}$
f) $y = \frac{3x-1}{2x+9}$

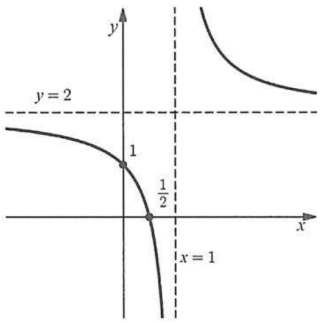
1.18 — Question 5.

For each of the following graphs, find the curve in the form

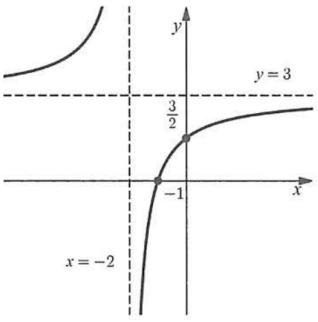
$$y = \frac{ax + b}{cx + d}$$

where a, b, c and d are integer constants to be found.

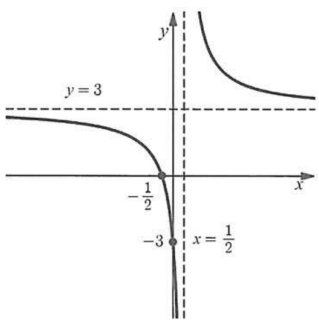
a)



b)



c)



1.18 — Question 6.

Sketch the following curves, where a and b are constants such that $a > 1$ and $b > \frac{1}{2}$ and $a > b$
Identify intersections with the axes and any asymptotes clearly.

a) $y = \frac{x+a}{x+1}$
b) $y = \frac{x-a}{x+2}$
c) $y = \frac{x-4}{x+a}$
d) $y = \frac{x+a}{x+b}$
e) $y = \frac{x+a}{x-b}$
f) $y = \frac{3x-a}{x-b}$
g) $y = \frac{x-a}{4x+b}$
h) $y = \frac{ax-1}{x-a}$

1.18 — Question 7.

Sketch each of the following pairs of equations on the same axes. Identify any intersections with the axes, any asymptotes and identify any points of intersection on the sketch.

a) $y = 4 \qquad y = \frac{x+1}{x-8}$
b) $y = x - 5 \qquad y = \frac{x+3}{x-6}$
c) $y = 8 - 2x \qquad y = \frac{x+5}{3x-5}$
d) $y = 5x - 34 \qquad y = \frac{x-2}{x-7}$
e) $y = 4 - x \qquad y = \frac{2x-8}{x+4}$
f) $y = 7x + 22 \qquad y = \frac{4-6x}{x+4}$
g) $y = 2x - 6 \qquad y = -\frac{3x+4}{2x+1}$

1.18 — Question 8.

A curve C_1 has equation $y = f(x)$ where

$$f(x) = \frac{2x+5}{7x-1}$$

a) Sketch the graph $y = f(x)$

A second curve, C_2 is a translation of curve C_1 by $\binom{1}{-2}$. C_2 is then transformed to C_3 by a stretch in the x direction scale factor 2

b) Find an equation for the curve C_2 in the form

$$y = \frac{ax+b}{cx+d}$$

where a, b, c and d are constants to be found.

c) Find an equation for the curve C_2 in the form

$$y = \frac{px+q}{rx+s}$$

where p, q, r and s are constants to be found.

1.18 — Question 9.

It is given that the graph of the rational function $y = f(x)$ has exactly two asymptotes; $y = 3$ and $x = -4$
It intersects the x-axis exactly once at $(2, 0)$

a) Find $f(x)$

b) Find the coordinates where the curve meets the y-axis.

c) Sketch $y = f(x)$

d) Find the value for which $f(x) = 6$

1.18 — Question 10.

It is given that $y = f(x)$ is a rational function.
Its graph has exactly two asymptotes, $y = 2$ and $x = 3$
It intersects the x-axis exactly once at $\left(\frac{3}{4}, 0\right)$

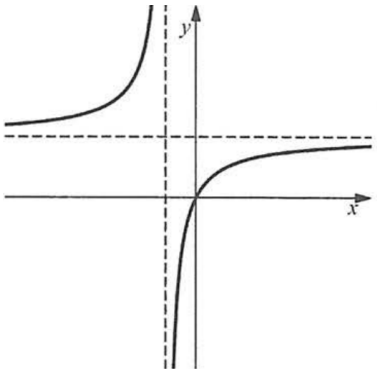
a) Find $f(x)$ in form $y = \frac{ax+b}{cx+d}$ where a, b, c and d are integers to be found.
b) Find the coordinates where the curve meets the y-axis.
c) Sketch $y = f(x)$
d) Find the coordinates when $f(x) = x + \frac{1}{2}$

1.18 — Question 11.

The curve C has the equation

$$y = \frac{2x}{ax + b}$$

where a and b are integers.



The asymptotes to the curve are $x = -\frac{1}{5}$ and $y = \frac{2}{5}$

a) Write down the value of a and b

It is known that the gradient at the origin is 2

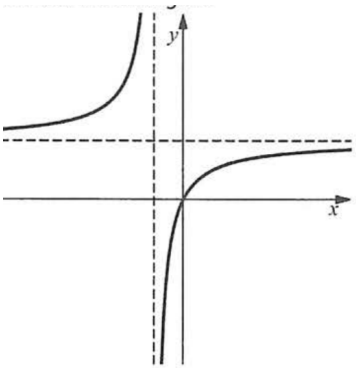
b) Write down the equation of the normal line to the curve at the origin.
c) Find the coordinates of the points where your answer to part b) meets curve C

1.18 — Question 12.

The curve C_1 has the equation

$$y = \frac{ax}{2x + b}$$

where a and b are integers.



The asymptotes to the curve are $x = -\frac{5}{2}$ and $y = \frac{7}{2}$

a) Write down the value of a and b

b) Show that the curve C_1 passes through the point $(1, 1)$

The curve C_2 has equation

$$y = -\frac{bx}{2x - a}$$

c) Show that the curve C_2 passes through the point $(1, 1)$

d) On a copy of the diagram above sketch C_2

e) Identify the relationship between C_1 and C_2

1.18 — Question 13.

The $y = f(x)$ has the equation

$$f(x) = \frac{ax}{bx + c}$$

where a, b and c are constants, and $\frac{a}{b}$ is a fraction in its simplest form. It has one point of intersection with the axes. The asymptotes to the curve are $x = 4$ and $y = \frac{5}{2}$

a) State the values of a, b and c

b) Explain why the condition $\frac{a}{b}$ is a fraction in its simplest form is required to be able to calculate the values of a, b and c

c) Sketch $y = f(x)$

It is known that $f'(0) = -\frac{1}{8}$

d) With reference to the graph, explain why there is exactly one root of the equation

$$\frac{ax}{bx + c} = -\frac{x}{8}$$

1.19 — Question 1. Find any asymptotes for each of the graphs of the curves below:

a) $y = \dfrac{x^2 - 1}{x^2 + 3}$

b) $y = \dfrac{x^2 - 16}{x^2 + 7}$

c) $y = \dfrac{x^2 + 3}{x^2 - 2}$

d) $y = \dfrac{3x^2 - 5x + 2}{x^2 + 3x}$

1.19 — Question 2. Sketch the following curves, identifying any intersections with the axes and any asymptotes clearly. It is given that each curve has a stationary point when $x = 0$

a) $y = \dfrac{x^2 - 4}{x^2 + 9}$

b) $y = \dfrac{x^2 - 9}{x^2 + 4}$

c) $y = \dfrac{x^2}{x^2 + 1}$

1.19 — Question 3. Solve the following equations:

a) $\dfrac{2x^2-3x+23}{2x^2+3x+9}=2$

b) $\dfrac{x^2-4x+2}{x^2-2x-5}=\dfrac{1}{2}$

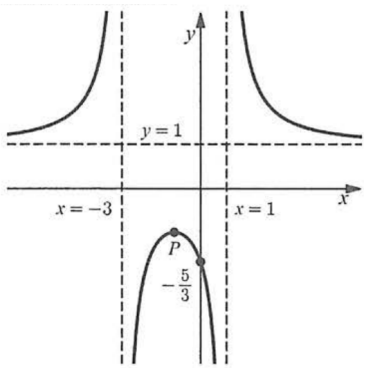
c) $\dfrac{x^2-4x+3}{x^2-3x+2}=\dfrac{4}{3}$

d) $\dfrac{x^2+4x-5}{x^2+4}=-\dfrac{1}{3}$

1.19 — Question 4. The curve

$$y=\frac{x^2+2x+5}{x^2+2x-3}$$

is shown in the diagram below. The local maximum is labelled P



a) By considering where the line $y=k$ intersects the curve, show that

$$(k-1)(k+1)\geq 0$$

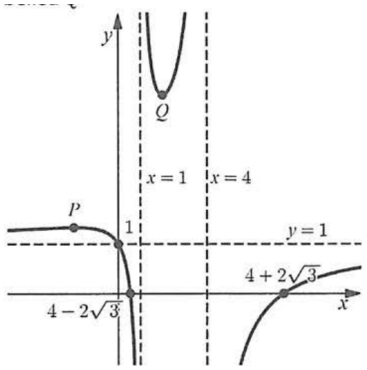
b) State the values that y can be.

c) Hence, find the y-coordinate of the point P

1.19 — Question 5. The curve

$$y = \frac{x^2 - 8x + 4}{x^2 - 5x + 4}$$

is shown in the diagram below. The local maximum is labelled P and the local minimum is labelled Q



a) By considering where the line $y = k$ intersects the curve, show that

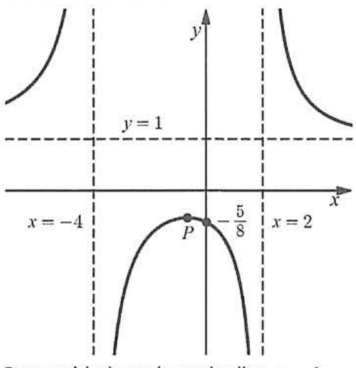
$$9k^2 - 48k + 48 \geq 0$$

b) State the values that y can be.
c) Hence, find the y-coordinate of the point P and the y-coordinate of the point Q

1.19 — Question 6. The curve

$$y = \frac{x^2 + x + 5}{x^2 + 2x - 8}$$

is shown in the diagram below. The local maximum is labelled P



a) By considering where the line $y = k$ intersects the curve, show that

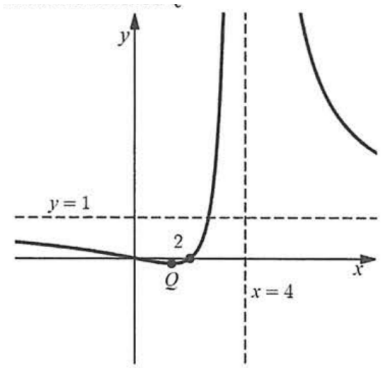
$$36k^2 - 16k - 19 \geq 0$$

b) State the values that y can be.
c) Hence, find the y-coordinate of the point P

1.19 — Question 7. The curve

$$y = \frac{x^2 - 2x}{x^2 - 8x + 16}$$

is shown in the diagram below. The local minimum is labelled Q



a) By considering where the line $y = k$ intersects the curve, show that

$$32k + 4 \geq 0$$

b) State the values that y can be.
c) Hence, find the y-coordinate of the point Q

1.19 — Question 8. A graph has equation

$$y = \frac{x + 2}{x^2 + 5}$$

a) Explain why the graph has no vertical asymptote.
b) Find the equation of the horizontal asymptote.
c) Show that, if the line $y = k$ intersects the graph, the x-coordinates of the points of intersection satisfy the equation

$$kx^2 - x + (5k - 2) = 0$$

d) Show that this equation has real roots if

$$-\frac{1}{10} \leq k \leq \frac{1}{2}$$

e) Hence, find the coordinates of the two stationary points on the curve.

1.19 — Question 9.

a) Prove that

$$-3 \leq \frac{12x}{x^2+4} \leq 3$$

b) Hence, sketch

$$\mathrm{f}(x) = \frac{12x}{x^2+4}$$

c) Given that $\mathrm{g}(x) = \dfrac{1}{\mathrm{f}(x)}$ explain why

$$-\frac{1}{3} \leq \mathrm{g}(x) < 0 \text{ and } 0 < \mathrm{g}(x) \leq \frac{1}{3}$$

1.19 — Question 10. Sketch each of the following curves on separate axes. Label any intersections with the axes, asymptotes and (local) maximum or (local) minimum points. You should find (local) maximum and (local) minimum points by considering where the line $y = k$ intersects the graph.

a) $y = \dfrac{x^2+4}{x^2-1}$

b) $y = \dfrac{x^2-4x+4}{x^2+4x}$

c) $y = \dfrac{x^2+8x+16}{x^2+2x+2}$

d) $y = \dfrac{x^2+4x}{x^2+2}$

e) $y = \dfrac{x^2-4x-5}{x^2+4x}$

f) $y = \dfrac{x^2-4x+2}{x^2-4x+3}$

1.19 — Question 11. Find any asymptotes for each of the graphs of the following curves:

a) $y = \dfrac{2x^2 + x + 1}{x^2 + 2x + 5}$

b) $y = \dfrac{3x^2 - 1}{x^2 + 2x + 3}$

c) $y = \dfrac{x^2 + 9}{4x^2 - 8x + 4}$

d) $y = \dfrac{5x^2 + 14x - 3}{2(2x^2 + 7x - 15)}$

1.19 — Question 12. A curve is given by the equation

$$y = \frac{ax^2 + bx + c}{x^2 - d}$$

where a, b, c and d are positive constants. Write down the equations of its asymptotes in terms of the constants.

1.19 — Question 13. A curve has equation

$$y = \frac{4x^2 + 4x + 11}{x^2 - x + 1}$$

It is known that the curve has a single maximum point and a single minimum point.

a) Write down the equation of the asymptote of the graph of the curve.

The line $y = k$ intersects the curve.

b) Show that

$$(k - 20)(3k - 8) \leq 0$$

c) Hence find the coordinates of the stationary points of the curve and identify which is the minimum and which is the maximum. Give a reason for your answer.

d) Hence, sketch the curve.

1.19 — Question 14. A curve has equation

$$y = \frac{2x^2 - 4x + 9}{x^2 - 4}$$

for $x \neq \pm p$ where p is a constant.

a) State the value of p

b) Write down the three equations of the asymptote of the graph of the curve.

The line $y = k$ intersects the curve.

c) Show that

$$(4k - 7)(k + 2) \geq 0$$

d) Hence find the coordinates of the stationary points of the curve and identify which is the local minimum and which is the local maximum.

e) Hence, sketch the curve.

1.19 — Question 15. A curve is given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x^2 - 8x + 1}{3x^2 + 1}$$

a) Write down the equation of the asymptote of the graph of the curve.

The line $y = k$ intersects the curve.

b) Show that

$$(3k + 5)(k - 3) \leq 0$$

c) Hence find the coordinates of the stationary points of the curve, stating which is the maximum and which is the minimum.

d) Sketch the graph of $y = \mathrm{f}(x)$

1.19 — Question 16. For each of the following, state the equation of the single asymptote of the graph and, without using Calculus, find the coordinates of the stationary point of the curve.

a) $y = \dfrac{1}{2x^2 + 3}$

b) $y = \dfrac{1}{3(x^2 + 2x + 2)}$

1.19 — Question 17. A curve is given by $y = \mathrm{f}(x)$ and it is known to have a local maximum at (a, b) where a and b are constants.
A second function is defined to be

$$\mathrm{g}(x) = \frac{1}{\mathrm{f}(x)}$$

It is given $\mathrm{f}(x)$ and $\mathrm{g}(x)$ have the same stationary point. State the type of stationary point and any restrictions on this occurring.

1.19 — Question 18. Two curves C_1 and C_2 are given by

$$\mathrm{f}(x) = x^2 - 4x + 5$$

and

$$y = \frac{1}{\mathrm{f}(x)}$$

respectively.

a) Find the minimum point on the graph of curve C_1
b) Sketch the graph of $y = \mathrm{f}(x)$
c) Without using Calculus, determine the single maximum point on the graph of curve C_2
d) Hence, state the equation of the line that is a tangent to both C_1 and C_2

1.19 — Question 19. A curve C_1 has equation $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{3x^2 - 8x + 7}{x^2 + 4x - 6}$$

The line $y = k$ intersects the curve C_1

a) Show that

$$(k + 1)(2k - 1) \geq 0$$

b) Hence find the coordinates of the stationary points of the curve C_1 and identify their type.

A second curve C_2 is given by

$$y = \frac{1}{\mathrm{f}(x)}$$

c) Show that the graph of C_2 has no vertical asymptotes.

d) State the equation of the line that is a tangent to both C_1 and C_2

1.19 — Question 20. A curve C is given by the equation

$$y = \frac{6 - x^2}{x^2 + kx + 2}$$

where k is a constant.

It is known that the graph of the curve only has one asymptote.

a) Write down the equation of the asymptote of the graph of the curve.

b) Find the possible values of k

c) Find where the graph of the curve C intersects the coordinate axes.

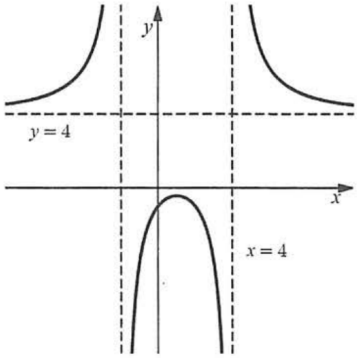
d) Sketch the graph of the curve given by

$$y = \frac{6 - x^2}{x^2 + 2}$$

1.19 — Question 21. The curve C has equation

$$y = \frac{ax^2 + 4bx + 2a}{x^2 + bx - 2a}$$

where a and b are integers and is shown in the diagram below.



One of the vertical asymptotes is $x = 4$
The horizontal asymptotes is $y = 4$

a) State the value of a
b) Find the value of b
c) State the equation of the second vertical asymptote of the curve.
d) Find the coordinate of the intercept of the curve with the y-axis.
e) Without using Calculus, find the coordinates of the local maximum point.

1.19 — Question 22. A curve is given $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{ax^2 + bx + c}{x^2 + px + q}$$

where a, b, c, p and q are integers.
It is known that:

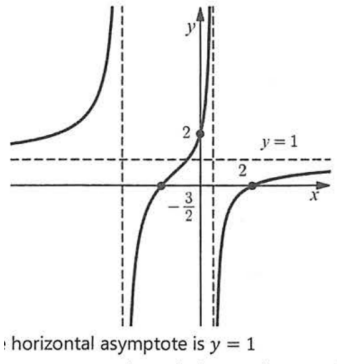
- There is a single asymptote at $y = 2$
- It crosses the y-axis at -1
- It crosses the x-axis at $-1 + \sqrt{\frac{3}{2}}$ and $-1 - \sqrt{\frac{3}{2}}$
- It has a minimum at $\left(-\frac{1}{2}, -2\right)$

a) Find $\mathrm{f}(x)$
b) Given that the curve also has a maximum, without using Calculus find the coordinates of this maximum point.
c) Sketch the graph of this function.

1.19 — Question 23. A curve is given $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{ax^2 + bx + c}{px^2 + qx + r}$$

where a, b, c, p, q and r are integers. The graph of $y = \mathrm{f}(x)$ is shown in the diagram below.



horizontal asymptote is $y = 1$

The horizontal asymptote is $y = 1$

The curve passes through the y-axis at $y = 2$ and the x-axis at $x = -\frac{3}{2}$ and $x = 2$

a) State the relationship between a and p

b) State the relationship between c and r

It is given additionally that $a = 2$ and that one of the vertical asymptotes is $x = -3$

c) Find $\mathrm{f}(x)$

d) Find the other vertical asymptote.

1.19 — Question 24. The curve C has equation

$$y = \frac{x^2 + p}{x^2 - q^2}$$

where p and q are positive integers.

Without using Calculus, find the range of values that y can take in terms of p and q

1.19 — Question 25. The curve C has equation

$$y = \frac{x^2 - px + p}{x^2 + q}$$

where p and q are positive integers.

a) State the asymptotes of curve C

b) Without using Calculus, show that the graph does not exist for

$$y < \frac{p + q - \sqrt{p^2 - 2pq + q^2 + p^2 q}}{2q}$$

and

$$y > \frac{p + q + \sqrt{p^2 - 2pq + q^2 + p^2 q}}{2q}$$

1.20 — Question 1. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x - 3}{x + 2}$$

a) State the asymptotes of the curve.

b) Sketch $y = \mathrm{f}(x)$

c) Hence, giving your answer in set notation, find the values when

$$\frac{x - 3}{x + 2} < 0$$

1.20 — Question 2. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{3x+4}{x-1}$$

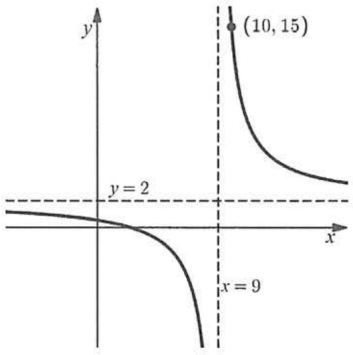
a) State the asymptotes of the curve.
b) Sketch $y = \mathrm{f}(x)$
c) Hence, giving your answer in interval notation, find the values when

$$\frac{3x+4}{x-1} \geq 0$$

1.20 — Question 3. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{2x-a}{x-b}$$

where a and b are positive integers is shown in the diagram below.



The asymptotes of the curve are given by the equations $x = 9$ and $y = 2$
a) State the values of a and b
b) Find the coordinates of where the curve meets the x-axis and y-axis.
c) Hence, find the values when

$$\frac{2x-a}{x-b} > 0$$

Give your answer in set notation.

1.20 — Question 4. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{1-3x}{4x+5}$$

a) State the asymptotes of the curve.
b) Sketch $y = \mathrm{f}(x)$
c) Hence, find the values when

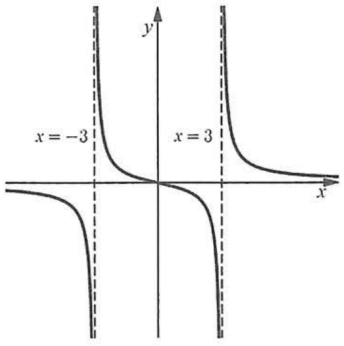
$$\frac{1-3x}{4x+5} < \frac{1}{2}$$

Give your answer in set notation.

1.20 — Question 5. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x}{x^2-k}$$

where k is a positive integer is shown in the diagram below.



The asymptotes of the curve are given by the equations $x = -3$, $x = 3$ and $y = 0$
a) State the value of k
b) For your value of k, solve

$$\frac{x}{x^2-k} = \frac{2}{5}$$

c) Hence, for your value of k find the values when

$$\frac{x}{x^2-k} \geq \frac{2}{5}$$

Give your answer in set notation.

1.20 — Question 6. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x^2 + x - 20}{x^2 + x - 6}$$

a) State the asymptotes of the curve.
b) Sketch $y = \mathrm{f}(x)$
c) Hence, find the values when

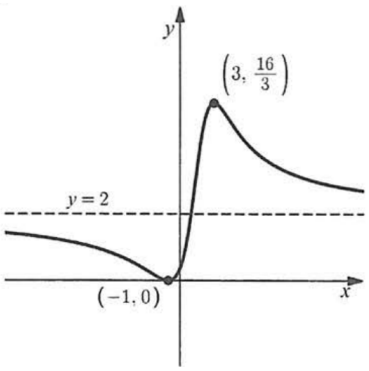
$$\frac{x^2 + x - 20}{x^2 + x - 6} \leq \frac{9}{2}$$

Give your answer in interval notation.

1.20 — Question 7. A curve is given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{ax^2 + 4x + b}{x^2 - 3x + c}$$

where a, b and c are integers is shown in the diagram below.



The asymptote of the curve is $y = 2$. There is a maximum at $(3, \frac{16}{3})$ and a minimum at $(-1, 0)$

a) Find the values of a, b and c
b) Hence, find the values when $\mathrm{f}(x) < 2$
Give your answer in interval notation.

1.20 — Question 8. The curve given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{x^2 - 1}{2x^2 - 4x + 3}$$

a) State any asymptotes of the curve.

The line $y = k$ intersects the curve.

b) Show that $(2k + 1)(k - 1) \leq 0$

c) Hence find the coordinates of the stationary points of the curve and identify their types.

d) Hence sketch $y = \mathrm{f}(x)$

e) Find the values of x when

$$\frac{x^2 - 1}{2x^2 - 4x + 3} \geq x - 1$$

Give your answer in set notation.

1.20 — Question 9. It is given that a rational function has equation $y = \mathrm{f}(x)$. Its graph has exactly two asymptotes $y = -2$ and $x = 1$

It intersects the x-axis exactly once at $(3, 0)$

a) Find $\mathrm{f}(x)$

b) Hence state the coordinates where the curve meets the x-axis and the y-axis.

c) Sketch $y = \mathrm{f}(x)$

d) Hence, find the value(s) for which $\mathrm{f}(x) \geq 4$

Give your answer in interval notation.

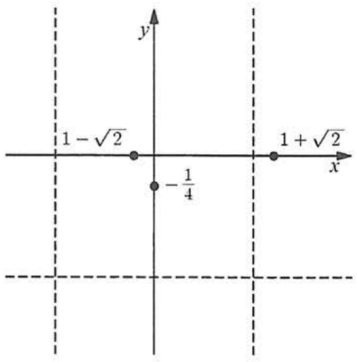
1.20 — Question 10. A curve is given by $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = \frac{3x^2 + ax + b}{x^2 + cx + 2}$$

where a, b and c are integers.
It has exactly one asymptote with the equation $y = 3$. The curve passes through the y-axis at -1. The curve has a maximum at $(3, \frac{29}{7})$ and a minimum at $(-1, -5)$

a) Find $\mathrm{f}(x)$
b) Find the values of x where the curve meets the x-axis.
c) Sketch $y = \mathrm{f}(x)$
d) Find the value(s) for which $\mathrm{f}(x) \geq -1$
Give your answer in set notation.

1.20 — Question 11. It is given that $y = \mathrm{f}(x)$ is a rational function of quadratic order in both the numerator and denominator of the fraction.
An incomplete sketch is shown below based on the following information:
Its graph has exactly three asymptotes which are given by equations $y = -1$, $x = 2$ and $x = -2$
It crosses the x-axis at $1 + \sqrt{2}$ and $1 - \sqrt{2}$ and it crosses the y-axis at $-\frac{1}{4}$



a) On a copy of the diagram above, complete the sketch of the curve.
You should label the asymptotes.
b) Find $\mathrm{f}(x)$
c) Hence, solve $\mathrm{f}(x) \leq -1$
Give your answer in interval notation.

1.21 — Question 1. Sketch each of the following curves:

a) $y^2 = 12x$

b) $y^2 = 2x$

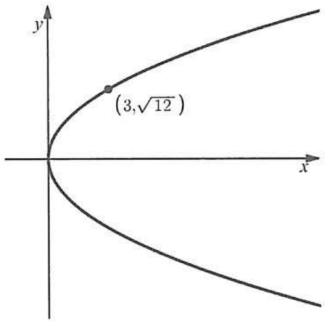
c) $y^2 = -6x$

1.21 — Question 2. Each of the following curves are of the form

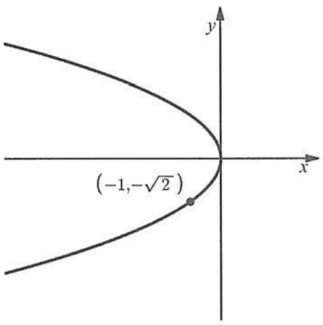
$$y^2 = ax$$

where a is a constant. In each case state the value of a

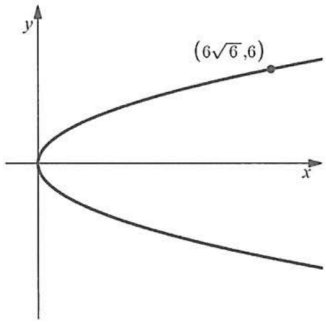
a)



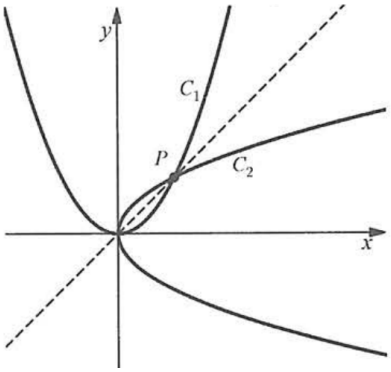
b)



c)



1.21 — Question 3. A curve C_1 is given by the equation $y = kx^2$ where k is constant. A second curve, C_2, is shown on the diagram which is the image of C_1 under a reflection over the line $y = x$



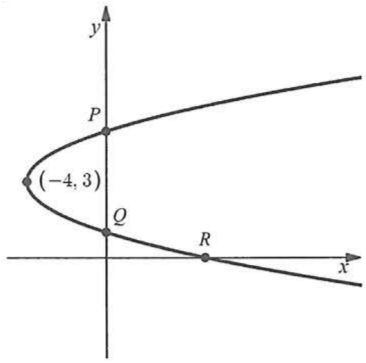
a) If $k = \frac{1}{4}$, find an equation for curve C_2 and the coordinates of the point P
b) If $k = 3$, find an equation for curve C_2 and the coordinates of the point P
c) If $k = a$, find an equation for curve C_2 and the coordinates of the point P

1.21 — Question 4. State the coordinates of the vertex for each of the parabolas below:
a) $y - 4 = (x - 2)^2$
b) $x - 4 = (y - 2)^2$
c) $y + 1 = (x - 5)^2$
d) $x + 1 = (y + 5)^2$

1.21 — Question 5. A parabola is given by the equation

$$(y-a)^2 = x - b$$

where a and b are constants is shown in the diagram below. The points of intersection with the y-axis are P and Q and with the x-axis is R



a) State the value of a and b
b) Find the coordinates of points P, Q and R

1.22 — Question 1. Sketch each of the following curves, labelled each intersection with the axes clearly.

a) $\dfrac{x^2}{9} + \dfrac{y^2}{25} = 1$
b) $\dfrac{x^2}{16} + \dfrac{y^2}{4} = 1$
c) $\dfrac{x^2}{2} + \dfrac{y^2}{5} = 1$
d) $x^2 + 9y^2 = 36$
e) $x^2 + 5y^2 = 5$
f) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ for constants a, b where $a > b$
g) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ for constants a, b where $a < b$
h) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ for constants a, b where $a = b$

1.22 — Question 2. Find the vertical tangents of each of the following curves:

a) $3x^2 + 5y^2 = 15$

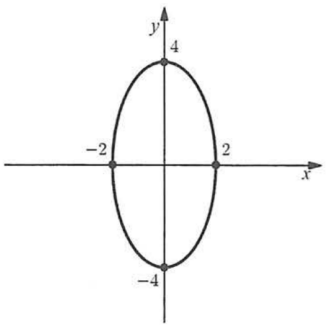
b) $5x^2 + 3y^2 = 15$

1.22 — Question 3. Each of the following curves are of the form

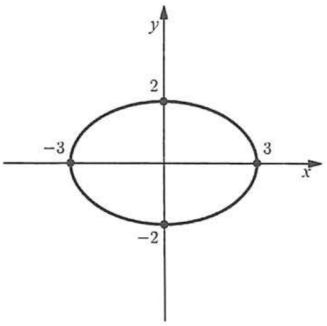
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where a and b are constants. In each case find the value of a and the value of b

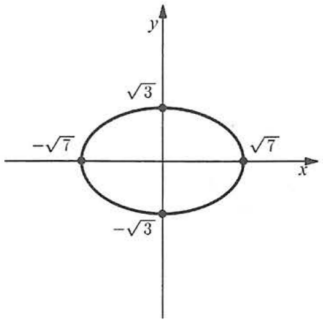
a)



b)



c)



1.22 — Question 4. For constants a and b, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

It passes through $P(1, 0)$ and $Q(0, -2)$

a) State the value of a and b

b) Sketch the curve including labelling any intersections with the axes.

1.22 — Question 5. For constants a and b, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

It passes through $P(-\sqrt{6}, 0)$ and $Q(0, 3)$

a) State the value of a and b

b) Sketch the curve including labelling any intersections with the axes.

1.22 — Question 6. For constants a and b, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{b} = 1$$

It passes through $P(2, \sqrt{6})$ and $Q(\sqrt{6}, -\sqrt{3})$

a) State the value of a and b

b) Sketch the curve including labelling any intersections with the axes.

1.22 — Question 7. For a constant a, an ellipse is given by

$$\frac{x^2}{a} + \frac{y^2}{2a} = 1$$

a) Find the points of intersection of the curve with the x-axis.

b) Find the points of intersection of the curve with the y-axis.

c) Sketch the curve including labelling any intersections with the axes in terms of a

d) Given the point $(2, -\sqrt{10})$ lies in the curve, find the value of a

1.23 — Question 1. State the asymptotes of each of the following curves:

a) $\frac{x^2}{16} - \frac{y^2}{4} = 1$

b) $x^2 - 9y^2 = 36$

c) $x^2 - 2y^2 = 16$

1.23 — Question 2. Sketch each of the following curves, label each intersection with the axes clearly and label any asymptotes in your sketch.

a) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

b) $\frac{x^2}{2} - \frac{y^2}{5} = 1$

c) $x^2 - 5y^2 = 5$

d) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for constants a, b where $a > b$

e) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for constants a, b where $a < b$

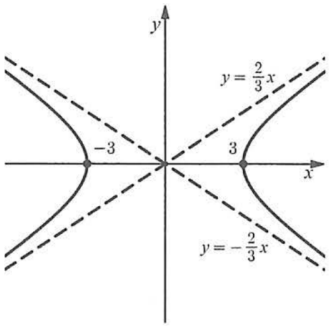
f) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ for constants a, b where $a = b$

1.23 — Question 3. For constants a and b, each of the curves below are of the form

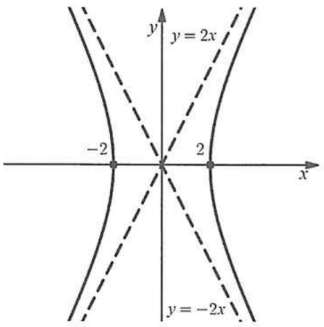
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

In each case find the value of a and b

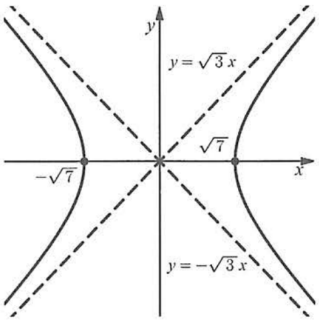
a)



b)



c)



1.23 — Question 4. For a constant a, a hyperbola is given by

$$\frac{x^2}{a^2} - \frac{y^2}{2a^2} = 1$$

a) Write down the equations of the asymptotes of the curve.
b) Hence, sketch the curve,

1.23 — Question 5. For constants a and b, a hyperbola is given by

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

It passes through $P(-3, 0)$ and $Q(6, \sqrt{6})$

a) Find the value of a and b

b) Hence, sketch the curve.

1.23 — Question 6. For constants a and b, a hyperbola is given by

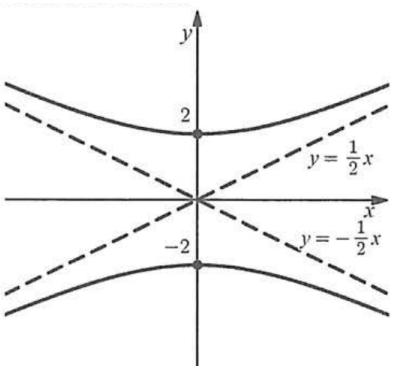
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

It passes through $P(2, -\sqrt{3})$ and $Q(-3, 3\sqrt{2})$

a) Find the value of a and b

b) Hence, sketch the curve.

1.23 — Question 7. Find the equation of the curve below in the form $x^2 - py^2 = q$ where p and q are constants to be found.



1.24 — Question 1. Sketch each of the following curves, in each case state asymptotes and any lines of symmetry:

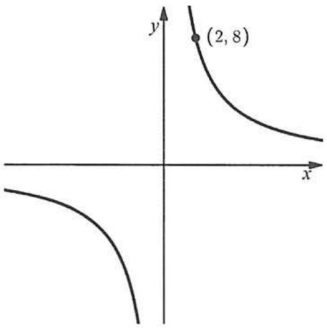
a) $xy = 9$
b) $xy = \frac{1}{4}$
c) $xy = -3$

1.24 — Question 2. Each of the following curves are of the form

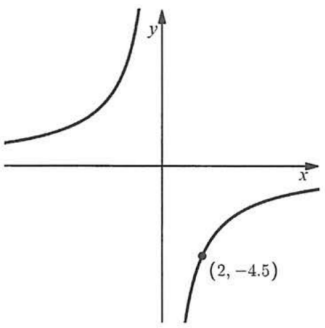
$$xy = a$$

where a is a constant. In each case state the value of a

a)



b)



1.24 — Question 3. Show that the two asymptotes of the curve

$$x^2 - y^2 = k$$

are perpendicular to each other.

1.24 — Question 4. Sketch each of the following curves, label each intersection with the axes clearly and include any asymptotes in your sketch.

a) $x^2 - y^2 = 1$

b) $x^2 - y^2 = 9$

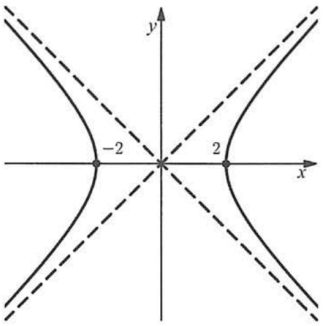
c) $\dfrac{x^2}{2} - \dfrac{y^2}{2} = 1$

d) $y^2 - x^2 = 3$

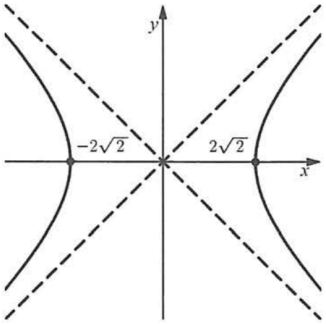
1.24 — Question 5. For curves below find the value of the constant k, given each may be written in the form

$$x^2 - y^2 = k$$

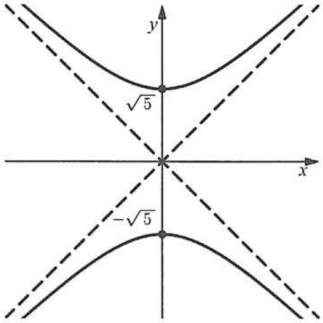
a)



b)



c)



1.24 — Question 6. For constant k, a curve given by the equation

$$x^2 - y^2 = k$$

The curve passes through $(a, -\sqrt{b})$ for positive constants, a and b

a) Find the value of k in terms of a and b

b) Find the values at which it passes through the x-axis in terms of a and b

1.25 — Question 1. Write down the equation of the image of the curve $y^2 = 4x$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor 3

b) A stretch, parallel to the y-axis, scale factor $\frac{1}{4}$

c) A translation by the vector $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$

d) A reflection in the x-axis.

e) A reflection in the line $y = x$

1.25 — Question 2. Write down the equation of the image of the curve $x^2 + 3y^2 = 3$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor $\frac{1}{2}$
b) A stretch, parallel to the y-axis, scale factor 5
c) A translation by the vector $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$
d) A reflection in the y-axis.
e) A reflection in the line $y = -x$

1.25 — Question 3. Write down the equation of the image of the curve $xy = 12$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor 3
b) A stretch, parallel to the y-axis, scale factor 2
c) A translation by the vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$
d) A reflection in the x-axis.
e) A reflection in the line $y = -x$

1.25 — Question 4. Write down the equation of the image of the curve given by $\frac{x^2}{4} - \frac{y^2}{5} = 1$ when it has been transformed by each of the following transformations:

a) A stretch, parallel to the x-axis, scale factor $\frac{1}{4}$
b) A stretch, parallel to the y-axis, scale factor 2
c) A translation by the vector $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$
d) A reflection in the x-axis.
e) A reflection in the line $y = x$

1.25 — Question 5. Each of the following curves are transformed by a stretch, parallel to the x-axis scale factor 2 Find the equation of the image and state the name of its shape:

a) $\frac{x^2}{3} + \frac{y^2}{16} = 1$
b) $\frac{x^2}{2} - \frac{y^2}{10} = 1$
c) $xy = 14$
d) $x^2 - y^2 = 4$

1.25 — Question 6. Each of the following curves are transformed by a stretch, parallel to the y-axis scale factor 3 Find the equation of the image and state the name of its shape:

a) $\frac{x^2}{18} + \frac{y^2}{2} = 1$
b) $\frac{x^2}{45} - \frac{y^2}{5} = 1$
c) $y^2 = -3x$
d) $x^2 - y^2 = 2$

1.25 — Question 7. A circle with equation $x^2 + y^2 = 4$ is transformed by the given transformation. In each case, determine the equation of the ellipse it forms in the form $\frac{x^2}{a} + \frac{y^2}{b} = 1$ where a and b are constants to be found.

a) A stretch, parallel to the x-axis, scale factor 2
b) A stretch, parallel to the y-axis, scale factor 2

1.25 — Question 8. Each of the following curves are reflected in the line $y = x$. In each case, find the equation of the image of the curve:

a) $y^2 = \frac{1}{2}x$
b) $3x^2 + 7y^2 = 21$
c) $xy = -6$
d) $\frac{x^2}{8} - \frac{y^2}{4} = 1$

1.25 — Question 9. Each of the following curves are reflected in the line $y = -x$. In each case, find the equation of the image of the curve:

a) $y^2 = \frac{1}{4}x$
b) $2x^2 + 5y^2 = 10$
c) $xy = \frac{1}{5}$
d) $\frac{x^2}{3} - \frac{y^2}{25} = 1$

Prerequisite knowledge for Q10-Q11: Matrix Transformations

1.25 — Question 10. The curve C_1 has equation $\frac{x^2}{4} + \frac{y^2}{8} = 1$

The matrix $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ maps C_1 onto C_2

a) Find the equation of the curve C_2

The matrix $\mathbf{N} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ maps C_1 onto C_3

b) Find the equation of the curve C_3

1.25 — Question 11. The curve C_1 has equation $\frac{x^2}{12} - \frac{y^2}{4} = 1$

The matrix $\mathbf{M} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ maps C_1 onto C_2

a) Find the equation of the curve C_2

b) Find the asymptotes of curve C_2

The matrix $\mathbf{N} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ maps C_1 onto C_3

c) Find the equation of the curve C_3

d) Find the asymptotes of curve C_3

1.25 — Question 12. Describe the transformation which maps:

a) $y^2 = 2x$ to $(y-3)^2 = 2x$

b) $\frac{x^2}{4} - \frac{y^2}{3} = 1$ to $\frac{x^2}{4} - \frac{(y+1)^2}{3} = 1$

c) $\frac{x^2}{9} + \frac{y^2}{16} = 1$ to $\frac{(x-1)^2}{9} + \frac{(y+3)^2}{16} = 1$

d) $\frac{x^2}{16} + \frac{y^2}{2} = 1$ to $\frac{x^2}{4} + \frac{y^2}{2} = 1$

e) $\frac{x^2}{2} - \frac{y^2}{5} = 1$ to $\frac{x^2}{18} - \frac{y^2}{5} = 1$

f) $\frac{x^2}{16} - y^2 = 4$ to $x^2 - 16(y+2)^2 = 64$

g) $\frac{x^2}{8} - \frac{y^2}{2} = 1$ to $(x+5)^2 - 4(y-1)^2 = 8$

h) $\frac{x^2}{16} - \frac{y^2}{2} = 1$ to $x^2 - 288y^2 = 16$

i) $\frac{x^2}{6} - \frac{y^2}{4} = 1$ to $2x^2 - 12x - 3y^2 - 18y = 21$

1.25 — Question 13. Sketch each of the following curves, label each intersection with the axes clearly:

a) $y^2 = 4(x+1)$

b) $(y-4)^2 = 2x$

c) $(y-3)^2 = 8(x+2)$

d) $y^2 = 2x - 6$

1.25 — Question 14. Sketch each of the following curves, label each intersection with the axes clearly and include any asymptotes in your sketch.

a) $x(y-2)=6$
b) $y(x+5)=-15$
c) $(x-2)(y+4)=12$
d) $3y+xy=9$
e) $xy-2y=5$

1.25 — Question 15. The curve C_1 given by equation $\frac{x^2}{25}+\frac{y^2}{5}=1$ is mapped to curve C_2 with equation

$$x^2+5(y+1)^2=25$$

a) Describe the transformation which maps curve C_1 to curve C_2
b) Find where the curve C_2 intersects with the coordinate axes.
c) Hence, sketch the curve C_2

1.25 — Question 16. A curve C is given by equation $\frac{x^2}{9} - \frac{y^2}{12} = 1$

a) State the type of curve C is.

b) State the asymptotes of C

c) Given that C is translated by $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$, find the asymptotes of the image of C

1.25 — Question 17. The curve C_1 has equation $\frac{x^2}{4} - \frac{y^2}{25} = 1$

a) Sketch the curve C_1

The curve C_1 is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ where k is a positive constant to give curve C_2

The curve C_2 passes through the origin.

b) State the value of k

c) Find the equations of the asymptotes of C_2 giving your answer in the form $y = mx + c$

1.25 — Question 18. The curve C_1 given by equation $\frac{x^2}{9} + y^2 = 1$ is mapped to curve C_2 with equation

$$x^2 - 4x + 9y^2 = 5$$

a) Find where the curve C_1 intersects with the coordinate axes.
b) Describe the transformation which maps curve C_1 to curve C_2
c) Find the coordinates of the maximum and minimum values of x and y on the curve C_2
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2

1.25 — Question 19. The curve C_1 given by equation $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is mapped to curve C_2 with equation

$$\frac{(x+4)^2}{9} - \frac{y^2}{4} = 1$$

a) State the equations of the asymptotes of curve C_1
b) Describe the transformation which maps curve C_1 to curve C_2
c) State the equations of the asymptotes of curve C_2
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2 labelling where the asymptotes intersect the axes.

1.25 — Question 20. The curve C_1 given by equation $\frac{x^2}{4} - \frac{(y-2)^2}{25} = 1$ is mapped to curve C_2 with equation

$$\frac{x^2}{4} - \frac{(y+3)^2}{25} = 1$$

a) Determine the equations of the asymptotes of curve C_1
b) Describe the transformation which maps curve C_1 to curve C_2
c) Find the asymptotes of the curve C_2
d) Find where the curve C_2 intersects with the coordinate axes.
e) Hence, sketch the curve C_2 labelling where the asymptotes intersect the axes.

1.25 — Question 21. The curve C_1 is given by the equation

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$

The curve C_2 is the image of C_1 after a translation, such that C_2 touches the x-axis at a single positive value of x and it touches the y-axis at a single positive value of y

a) Find the vector that translates C_1 to C_2
b) State the points of intersection of C_1 and C_2
c) Sketch both C_1 and C_2 on the same axes.
d) Find the gradient m of the equations of the tangent lines which are a tangent line to both C_1 and C_2
e) Show that for your m the tangent lines are $y = mx + 5\sqrt{2}$ and $y = mx - 5\sqrt{2}$

1.25 — Question 22. An ellipse E is given by the equation

$$\frac{x^2}{8} + \frac{y^2}{3} = 1$$

A straight line L is given by the equation

$$y = 2x + 1$$

a) Write down the coordinates of the points where the ellipse E intersects the coordinate axes.

The ellipse E is translated by the vector $\begin{pmatrix} k \\ 0 \end{pmatrix}$ where k is a positive constant.

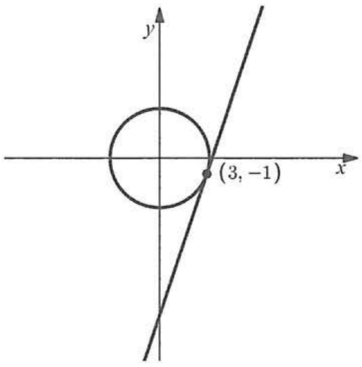
b) Write down the equation of the translated ellipse.

c) Show, that if the translated ellipse intersects the line L then the points of intersection must satisfy the equation

$$35x^2 + (32 - 6k)x + (3k^2 - 16) = 0$$

d) Given that the line L is a tangent to the translated ellipse, find the value of k

1.25 — Question 23. The diagram shows a circle C and a line L, which is the tangent to C at the point $(3, -1)$



The equations of C and L_1 are

$$x^2 + y^2 = 10 \text{ and } 3x - y = 10$$

respectively.

The circle C is now transformed by a stretch with scale factor 3 parallel to the x-axis. The image of C under this stretch is an ellipse E. The line is also transformed by a stretch with scale factor 3 parallel to the x-axis to form line L_2

a) Find the equation of the ellipse E

b) Find the equation of the line L_2

c) Show that L_2 is a tangent to the ellipse E and hence find the point of intersection.

d) Sketch the ellipse E and the line L_2

1.26 — Question 1. The curve C and the line L have equations $y^2 = 12x$ and $y = 6 - 4x$ respectively.

a) Sketch C and L on the same axes.

b) Find the points of intersection between the curve C and the line L

c) Find the exact length of the line segment between the two points of intersection.

1.26 — Question 2. Two curves C_1 and C_2 have equations $y^2 = 40x$ and $\frac{x^2}{4} + \frac{y^2}{9} = 1$ respectively.

a) Sketch C_1 and C_2 on the same axes.

b) Find the points of intersection between the curve C_1 and the curve C_2

1.26 — Question 3. Two curves C_1 and C_2 have equations $y^2 = -2x$ and $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively.

a) Sketch C_1 and C_2 on the same axes.

b) Find the points of intersection between the curve C_1 and the curve C_2

1.26 — Question 4. A curve C and the line L have equations $xy = 16$ and $y = -kx + 8\sqrt{2}$ respectively.

a) Sketch C

b) Find the x-values of the points of intersection between the curve C and the line L in terms of k

c) Find the value of k such that $y = -kx + 8\sqrt{2}$ is a tangent to the curve.

Prerequisite knowledge for Q5-Q6: Volume of Revolution

1.26 — Question 5. A hyperbola is given by equation

$$\frac{x^2}{a^2} - \frac{y^2}{3a^2} = 1$$

where a is a positive constant.

a) Write down the equations of the asymptotes of the curve in terms of a

b) Sketch the curve.

The finite region bounded by the curve, the positive x-axis, the line $x = a$ and the line $x = 2a$ is rotated 360° about the x-axis.

c) Find the volume of the solid generated in terms of a, giving your answer in exact form.

1.26 — Question 6. An ellipse is given by equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

a) Given that $a > b$, sketch the curve.

The finite region bounded by the curve, the positive x-axis and the positive y-axis is rotated 360° about the y-axis.

b) Find the volume of the solid generated in terms, giving your answer in exact form.

1.26 — Question 7. A parabola P has equation

$$y^2 = 12(x - k)$$

where $k > 0$
The line $y = mx$ is a tangent to P
Without using Calculus, prove that $m = \pm\sqrt{\frac{3}{k}}$

1.26 — Question 8. An ellipse E has equation

$$\frac{x^2}{4} + \frac{y^2}{16} = 1$$

The line with equation $y = mx + 5$ is a tangent to E. Without using Calculus, find the possible values of m

1.26 — Question 9. An ellipse E_1 is given by the equation

$$(x-5)^2 + \frac{y^2}{9} = 1$$

A second ellipse E_2 is given by the equation

$$\frac{x^2}{9} + (y-5)^2 = 1$$

a) Sketch E_1 and E_2 on the same axes.

A tangent line of E_1 is of the form $x + y = k$

b) Find the possible values of k

c) Show that this tangent line is also a tangent to E_2

1.26 — Question 10. Find the equation of the locus of points such that the distance from the point $(0, -4)$ is always equal to the distance from the line $y = 2$, giving your answer in the form

$$x^2 = p(y+q)$$

where p and q are constants to be found.

1.26 — Question 11. Find the equation of the locus of points such that the distance from the point $(9, 0)$ is always three times the distance from the line $x = 1$